

Unbalanced Mach–Zehnder Interferometers for Group-Index Extraction in TE and TM Silicon Strip Waveguides

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Abstract—Unbalanced Mach–Zehnder interferometers (MZIs) were designed to extract the group index n_g of a 500×220 nm silicon strip waveguide from the free spectral range (FSR). Simulated values at 1550 nm are $n_g = 4.196$ (TE) and 3.744 (TM). TE and TM devices are included so that the measured n_g values can be used to estimate the fabricated waveguide width and thickness.

I. INTRODUCTION

The objective is to extract the waveguide group index from the FSR of unbalanced MZIs, for TE and TM, and to use both values to estimate the fabricated waveguide geometry.

II. THEORY

A. MZI Transfer Function

Following [1]:

$$\frac{I_o}{I_i} = \frac{1}{4} \left| e^{-j\beta_1 L_1 - \frac{\alpha}{2} L_1} + e^{-j\beta_2 L_2 - \frac{\alpha}{2} L_2} \right|^2, \quad \beta = \frac{2\pi n_{\text{eff}}(\lambda)}{\lambda}. \quad (1)$$

For identical, lossless arms:

$$\frac{I_o}{I_i} = \frac{1}{2} [1 + \cos(\beta \Delta L)], \quad \Delta L = L_2 - L_1. \quad (2)$$

B. Group Index from the FSR

The phase difference between the arms is $\phi = 2\pi n_{\text{eff}} \Delta L / \lambda$. Adjacent peaks are separated by $\Delta\phi = 2\pi$, so $|d\phi/d\lambda| \text{FSR} = 2\pi$, with

$$\frac{d\phi}{d\lambda} = -\frac{2\pi \Delta L}{\lambda^2} \underbrace{\left(n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda} \right)}_{n_g} \Rightarrow n_g = \frac{\lambda^2}{\Delta L \cdot \text{FSR}}. \quad (3)$$

III. MODELLING AND SIMULATION

A. Waveguide

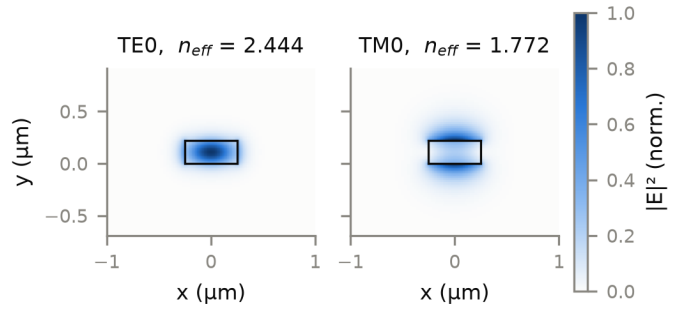


Fig. 1. TE0 and TM0 mode intensity at 1550 nm.

Silicon strip waveguide, 500×220 nm, SiO_2 cladding, quasi-TE and quasi-TM polarizations. Modes were simulated with Lumerical MODE (FDE) [2] (Fig. 1).

B. Effective and Group Index

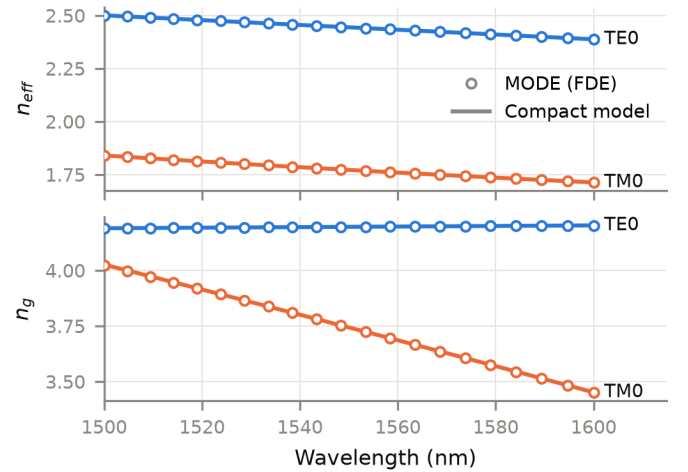


Fig. 2. Effective index (top) and group index (bottom). Markers: MODE; lines: compact model.

The compact model (Fig. 2), with λ in μm and $\lambda_0 = 1.55 \mu\text{m}$:

$$n_{\text{eff,TE}}(\lambda) = 2.4442 - 1.1302(\lambda - \lambda_0) - 0.0414(\lambda - \lambda_0)^2, \quad (4)$$

$$n_{\text{eff,TM}}(\lambda) = 1.7719 - 1.2721(\lambda - \lambda_0) + 1.8526(\lambda - \lambda_0)^2. \quad (5)$$

With $n_g = n_{\text{eff}} - \lambda dn_{\text{eff}}/d\lambda$: $n_{g,\text{TE}} = 4.196$ and $n_{g,\text{TM}} = 3.744$ at 1550 nm .

C. Design Variations

TABLE I
DESIGNS AND SIMULATED PERFORMANCE AT 1550 nm

Device	Pol.	ΔL (μm)	FSR (nm)	n_g
Loopback	TE	—	—	—
MZI1	TE	50	11.451	4.1960
MZI2	TE	100	5.726	4.1960
MZI2b (copy)	TE	100	5.726	4.1960
MZI3	TE	200	2.863	4.1960
Loopback	TM	—	—	—
MZI-TM	TM	100	6.418	3.7437

ΔL is chosen so that the FSR is below the 50 nm measurement bandwidth, with at least three fringes in the grating-coupler passband (Table I). Loopbacks are used to remove the grating-coupler response.

D. Circuit Simulation

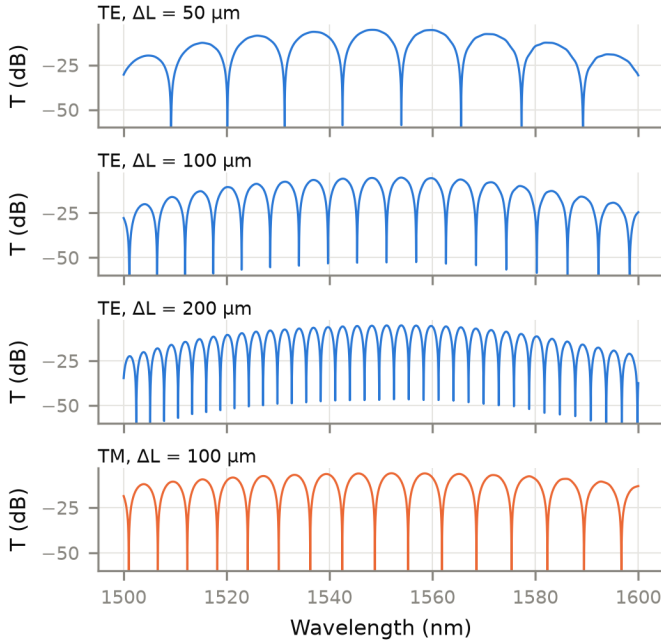


Fig. 3. Simulated MZI transmission.

Circuits were simulated in Lumerical INTERCONNECT [2] using the EBeam library [3] grating couplers (TE and TM) and Y-branch (Fig. 3). The arms use the compact model of Section III-B. The EBeam library waveguide model gives $n_g = 4.191$ (TE) and 3.730 (TM), within 0.4% of the MODE values.

E. Group-Index Extraction

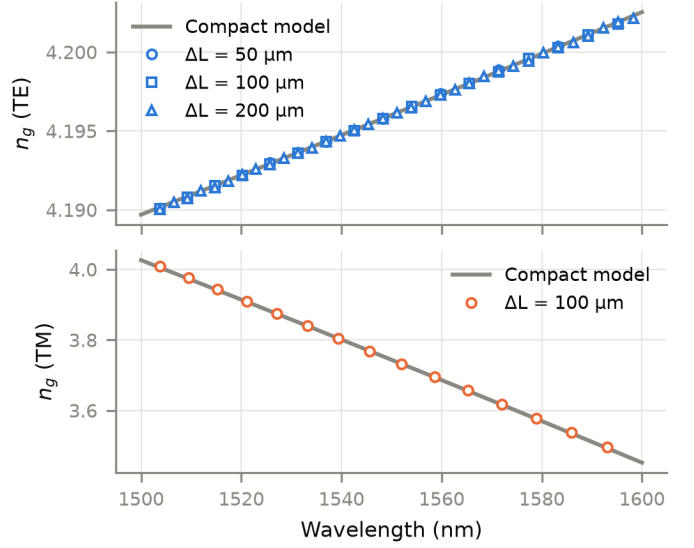


Fig. 4. n_g extracted from the simulated FSR (markers) and compact model (line).

Applying Eq. (3) to the simulated spectra gives the n_g values in Table I and Fig. 4.

F. Fabrication Sensitivity

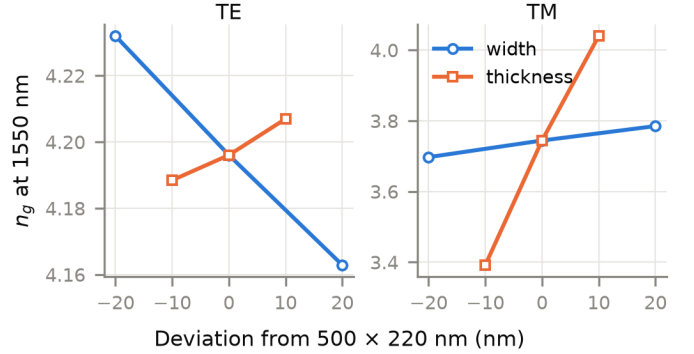


Fig. 5. n_g versus width and thickness deviation.

TABLE II
 n_g SENSITIVITIES (NM^{-1}) AT $500 \times 220 \text{ nm}$, 1550 nm

	$\partial n_g / \partial w$	$\partial n_g / \partial h$
TE	$a_w = -1.73 \times 10^{-3}$	$a_h = +0.93 \times 10^{-3}$
TM	$b_w = +2.20 \times 10^{-3}$	$b_h = +3.24 \times 10^{-2}$

n_g at 1550 nm was simulated in MODE for widths of $480\text{--}520 \text{ nm}$ and thicknesses of $210\text{--}230 \text{ nm}$ (Fig. 5, Table II). TE n_g depends mainly on width, and TM n_g mainly on thickness.

G. Data Analysis

- 1) Subtract the loopback spectrum (dB) to remove the grating-coupler response.

- 2) Locate the transmission minima.
- 3) From adjacent minima, compute the FSR and $n_g(\lambda)$ with Eq. (3).
- 4) Take n_g at 1550 nm; the spread over the TE devices gives the uncertainty.
- 5) Using the sensitivities of Table II,

$$\Delta n_{g,\text{TE}} = a_w \Delta w + a_h \Delta h,$$

$$\Delta n_{g,\text{TM}} = b_w \Delta w + b_h \Delta h,$$

solve for the fabricated width and thickness deviations Δw , Δh .

IV. FABRICATION

V. EXPERIMENTAL DATA

VI. ANALYSIS

VII. CONCLUSION

REFERENCES

- [1] L. Chrostowski and M. Hochberg, *Silicon Photonics Design: From Devices to Systems*. Cambridge University Press, 2015.
- [2] Ansys Lumerical MODE and INTERCONNECT. <https://www.ansys.com/products/optics>
- [3] SiEPIC EBeam PDK. https://github.com/SiEPIC/SiEPIC_EBeam_PDK